

Professional Summary

AI/ML Engineer with hands-on experience building production-ready **machine learning and deep learning systems** using PyTorch, TensorFlow, and Scikit-Learn. Developed real-time prediction APIs, LLM-powered applications, and computer vision models. Strong background in **model optimization, feature engineering, and deployment**. Interested in building scalable AI systems for real-world applications.

Education

Bachelor of Technology

Major: Electrical Engineering

Minor: Artificial Intelligence

CGPA: 7.45

2022 – 2026

Technical Skills

Programming

Python, C++, SQL

ML/DL Frameworks

TensorFlow, Scikit-Learn, PyTorch

AI Specializations

Deep Learning, Natural Language Processing, Computer Vision, Time-Series Forecasting

LLM Stack

LangChain, Retrieval-Augmented Generation (RAG), OpenAI API, HuggingFace

Tools

Git, FastAPI, OpenCV, Power BI

Concepts

Feature Engineering, Model Optimization, Hyperparameter Tuning, Data Preprocessing

Relevant Machine Learning Coursework

- Data Structures
 - Machine Learning
 - Pattern Recognition
 - Fundamentals of Artificial Intelligence
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Professional Experience

AI/ML Developer Intern

Mar 2024 – Aug 2024

- Built a **real-time cryptocurrency prediction API** using LSTM time-series models in Python.
- Improved model accuracy by **15%** through feature engineering and hyperparameter tuning.

- Designed data pipelines for preprocessing financial time-series datasets.
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AI/ML Engineer Intern

May 2025 – Aug 2025

- Developed an **LLM-powered search engine** using LangChain and OpenAI APIs.
 - Implemented query understanding and document retrieval pipeline using **RAG architecture**.
 - Optimized inference cost and response latency for production deployment.
 - Collaborated with product teams to improve chat-based user experience and engagement.
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Research Intern

May 2025 – Oct 2025

- Built optimization pipelines using **Adam + L-BFGS hybrid optimization in PyTorch**.
 - Extracted metrics such as **Q-factor, extinction ratio, and spectral contrast** from resonance spectra.
 - Modeled non-Lorentzian photonic responses for sensing applications.
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Projects

Marine Plastic Detection System

- Built a **CNN-based computer vision model** for marine plastic detection.
- Developed an image classification pipeline integrated with a robotic prototype for environmental monitoring.

Cryptocurrency Price Prediction

- Developed an **LSTM deep learning model** for cryptocurrency price prediction using time-series data.
 - Improved prediction accuracy through feature engineering and parameter tuning.
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Achievements

- **First Runner-Up** in a national hackathon for developing a live parking system.
 - **3rd Place among 150+ teams** in a technology competition with a Web3-based solution.
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Additional Information

Languages: English, Hindi

Certifications:

- Fundamentals of Artificial Intelligence
- Mentorship Program Certification